Spring '24 – User Research Methods – Professor Charles Raffaele
Observation Design Document REVISION/REFLECTION – Darryl Hutchinson

REFLECTION INTRO

I originally set up this observation to get a better look at how inexperienced people would be able to navigate through a game environment. I did what I could do incorporate what I know about game design as well as what I've learned thus far in this course. I thought I had a lot of valuable perspective, but I still encountered so many surprises and unexpected insights that I could write an entirely new observation protocol. However, the point of this reflection is to revise so I will just focus on what mistakes I made, what minor changes I can make to this design, and what alternate directions I might go in the future.

MISTAKES / ASSUMPTIONS

A. To start, I made some setup mistakes. I originally had the respondent filling out the pre-test questions from a separate chair with no computers within reach. That was good to keep them focused on the task, but one of the questions was inadvertently leading. Question 5 asked "when have you played a video game most recently?" The potential answer were, within the last two days/week/month/year. I didn't provide an option for time period over a year, but the respondent wrote in "(x) over one year" on their own.
B. While the respondent was filling out the pre-test, I was organizing the test setup and was able to confirm that the game demo was running and I thought my documents were all prepared. However, after starting the test I realized I was missing a page of questions.
C. Once the test started, the respondent held the controller in a way I did not expect. And made a lot of gameplay decisions that I did not foresee. I think I did a good job of holding my tongue and not guiding the player in any particular direction. However, there were so many surprises that my observation form wasn't prepared to capture and my note taking could barely keep up with to fill in the gaps so I was scrambling to keep pace with everything I was seeing.
D. During gameplay, I expected the player would see specific landmarks in the game environment and without specific prompts, be able to follow them and make progress. Therefore, I organized my observation form around how much time would be taken between those points of interest. However, the participant did so many other interesting things in between each POI and I didn't have a good code setup ahead time for account for these actions, so I ended up scrambling to try and capture what I could with hand written notes which I found to be very inefficient.
    a. The respondent also focused on / was more interested in / struggled with many things that I did not anticipate.
E. The respondent was taking far longer than anticipated with pre-test, gameplay test and post-test. Getting out of the starting cave took about 5 minutes and I was planning for the entire test to take 10 minutes. Despite taking longer than I anticipated, the respondent was smiling the entire play session. I also hadn't yet observed the initial target of time/actions between points of interest. Unsure of how best to proceed, I decided to let the session run overtime to get the observation I was targeting and to allow the respondent to continue enjoying the exploration of the game world.

ADJUSTMENTS

A. I will change that question to reflect time periods over one year. I can also review other questions for potentially leading question/answer wording.

B. I was able to ask the missing questions at the end of the session, but I should do better to make sure the entire setup and all documents are ready to go before the respondent even arrives. Maybe I could even sample test myself, just to make sure all parts of the test are ready.

C. My research questions perhaps did not match the respondent's experience level so I could refine them better capture what I'm trying to observer. Also, a video recording that I could refer back would also help with capturing unexpected phenomena. Being in a separate room, making voice notes might also be helpful. I basically don't think written memos are good for me to use.

D. I would refine my observation form with slightly more open ended labels. I could also have more shorthand codes ready to capture unexpected phenomena.

E. If I were stricter with timing, I wouldn't have observed what I was targeting. And if I were more open with questions, I would have been better prepared to capture alternate behaviors.

USER FEEDBACK / ANALYSES

Observation surprises:

- Respondent used the controller/left controller on table in a position I hadn't anticipated
- Respondent found in-game text to proceed to quickly
- Responded tried to open doors and enter windows all of which were not intractable
- Respondent read one in-game prompt to explore the island, and thus did not understand the objective of collecting the emeralds until one was touched and the on screen prompt said 1/7
- Respondent tried to interact with in-game wildlife, but the game did not provide much possible interaction
- Respondent was smiling the entire play session
- Respondent took longer than expected but was able to progress without researcher's direction
    - Researcher only explained player/camera movement via joystick and jump via face button
- Respondent was thinking aloud for roughly 15~25 % of gameplay without being prompted to do so. This heavily impacted my note taking.

USER FEEDBACK CONCLUSION

Based on much of the respondent's feedback, I encountered many things I did not expect. The way the respondent held the controller, focused on certain things in the environment, did not use the jump button much, as well as various verbal commentary made throughout the play session all combined to change my perspective on this entire observation. I've listed some of the adjustments I could make above that could improve this observation design. However, given the time, I would now much rather redesign this entire observation from the ground up with different foundational questions. For example, I would like to better understand how the game teaches the player its rules. During this gameplay observation session I realized only some of the many assumptions I have as an experienced gamer that my inexperienced respondent didn't have any reference for. I also noticed that despite the game being mechanically simple, the design of the game itself also makes many of these assumptions. So a primary question for a new observation might be "how/where/in what ways does the game teach the player the rules of the game world?"

INTRODUCTION & RESEARCH QUESTIONS

Though they have great potential for entertainment and learning, video games are typically very complex and it can be easy for inexperienced players to get overwhelmed and lose site of the objectives the designer wants them to experience. To better understand intuitive level design and which game elements do the best job of capturing player attention regardless of experience level such that they don't get lost and can progress through an experience, I want to analyze the upcoming video game, Europa.

Designed and developed by Helder Pinto and Novadust studios, Europa is a bright and colorful Sci-Fi adventure game with a heavy influence of painterly Ghibli style visual animation, smooth freeform movement and traversal with low stress exploration. The game has been in development since 2017 and is currently available in demo form with a scheduled release window of spring 2024.

I am interested in exploring pacing/pathing and environmental storytelling. I would like to know specifically how game elements in the game world can communicate which ways players can travel without using text or unnatural UI markers. Since video games are relatively complex bundles of systems that are easy to get lost within, I want to try and understand how to better lead the player along an experience with natural elements that help maintain immersion.

The target audience for this research could be Novadust studios themselves to gain more perspective on what their players are experiencing. With more feedback about what is drawing player attention, adjustments can be made to the level design to further avoid players getting lost or confused in the game. Another audience could also be other developers and/or researchers who want to more deeply understand how specific elements in Europa create specific responses or perceptions in players. Using this analysis as a case-study towards some generalizable level design techniques may reveal applications to other future projects.

The results of this analysis will be submitted to Professor Raffaele for evaluation.

Research Questions:

1. What path does the player take through this open game environment?
    a. So we can understand if the critical path is clear and understood for the player.
2. Where does the player spend the most time?
    a. So we can understand what points the player finds interesting.
3. How does the player spend time in each area?
    a. So we can understand what the player enjoys or is trying to do in the game world.

STUDY DESIGN

Observation is an appropriate approach for the aforementioned research questions and intended analysis in order to see what actions players naturally engage with and which directions they choose without solicitation or guidance. Rather than asking participants to self-report on what they see during the experience (which will break their immersion and negatively affect the observation data) or what they can recall after going through the experience (relying too heavily on memory), it will be more clear and direct to observe players as they are making authentic traversal decisions and executing actions in-

game in real time. I want to see if and how the design of game world itself communicates to the player where it wants them to go. In addition to observation data, a brief pre-game questionnaire and semi-structured post-interview may help establish participant background and fill in gaps of interpretation, respectively, that may be unclear if only considering observation data.

Furthermore, if I had more access, I would ideally have two treatment groups that each get a largely similar build of the game with one significant difference either in colors, landmarks, audio cues or other signals to try and determine which ones are actually most effective in communicating different things to the player. However, for this analysis, the participant (and future participants relative to this study) will get the same treatment because I only have access to one build of the game.

Variables to be measured will include:

- RQ1/RQ2 - Time spent at points of interest
- RQ1/RQ2 - Time spent between points of interest,
- RQ3 - What do players do when they reach a point of interest
- RQ3 - What do players do when they are moving between points of interest

*Additional variables to be measured if provided access to debug tools:

- RQ1/RQ2 - Vector map of the path player takes through the game
- RQ1/RQ2 - Number of divergences from the critical path
- RQ3 - Degree of divergences from the critical path

Quantitative time data (time spent), activity data (actions executed) and, if possible, vector data (paths taken via debug tools) can be collected by researcher for each respective research question. Additionally, qualitative survey data will be collected by research moderator via pre-game questionnaire and semi-structured post-interview. The data will be collected either at researcher's home, 370 Jay Street, or another mutually convenient location for the participant and researcher.

Since recruitment efforts are ongoing, a convenient test location has not yet been confirmed.

Instruments will include:

- Learner profile gaming experience pre-questionnaire
- User Observation data/video
    - Observation form
- Semi-structured gaming experience post-interview
- **User Position Logs* (**if provided Debug access**)

PARTICIPANT PROFILE

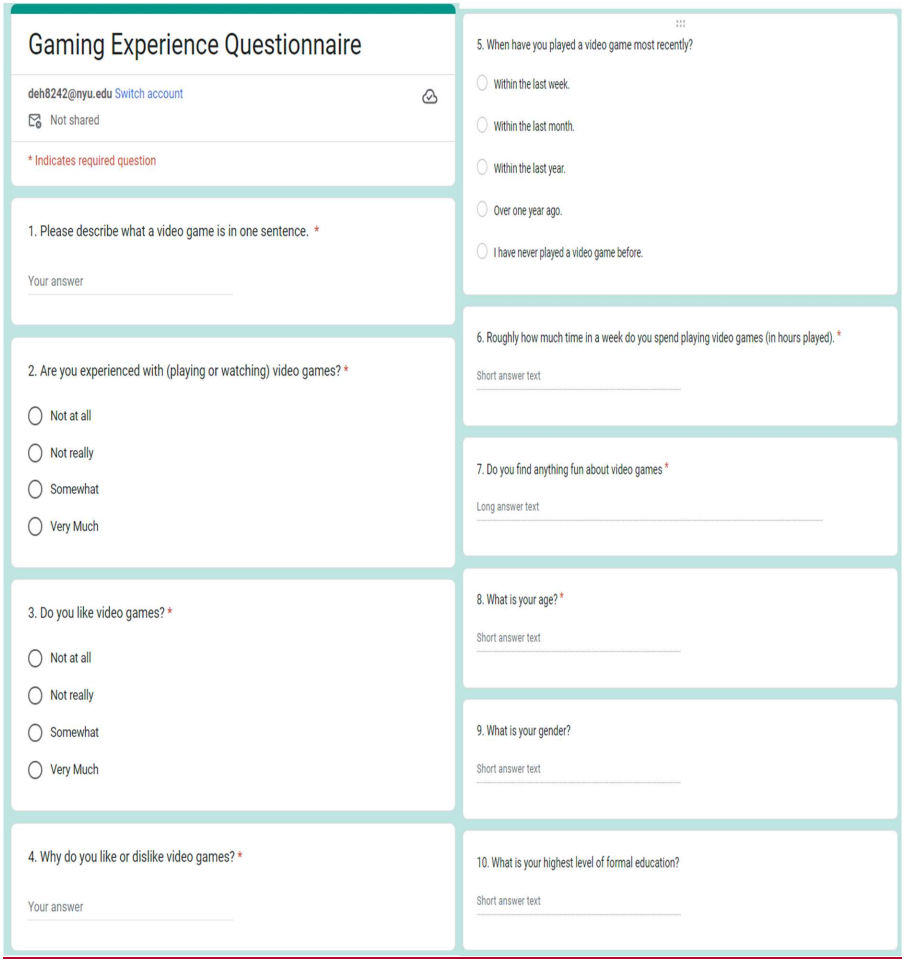
Data will be collected from a group size of one classmate and/or one family member that has little to no experience with playing and/or watching video games. ~~Recruitment efforts are currently still ongoing.~~

~~Placeholder Persona~~

~~I found this participant through class. She is a graduate level student in the Learning Technology and Experience Design program. She is 27 years old, identifies as an American middle-class woman who works as a dentist's assistant for a living and is not a gamer. She claims not to play video games on any regular basis and hasn't actually picked up a game controller since she was in elementary playing with school friends. She enjoys reggae music, sci-fi books and long walks on the beach in her free time.~~

This participant is a personal relationship. They are very inexperience with video games and is 70 years old. He identifies as male and his highest level of formal education is college.

INSTRUMENTS & MEASURES



Gaming Experience Questionnaire

deh8242@nyu.edu Switch account

Not shared

* Indicates required question

1. Please describe what a video game is in one sentence. *

Your answer

2. Are you experienced with (playing or watching) video games? *

- Not at all
- Not really
- Somewhat
- Very Much

3. Do you like video games? *

- Not at all
- Not really
- Somewhat
- Very Much

4. Why do you like or dislike video games? *

Your answer

5. When have you played a video game most recently?

- Within the last week.
- Within the last month.
- Within the last year.
- Over one year ago.
- I have never played a video game before.

6. Roughly how much time in a week do you spend playing video games (in hours played). *

Short answer text

7. Do you find anything fun about video games *

Long answer text

8. What is your age? *

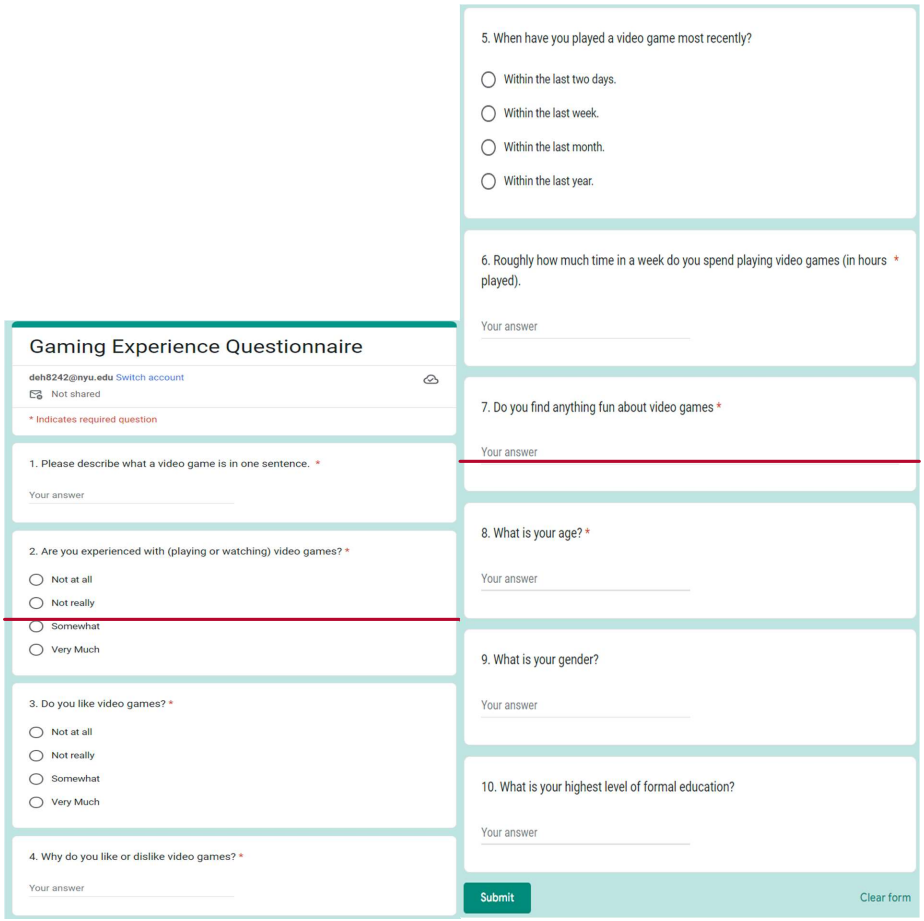
Short answer text

9. What is your gender?

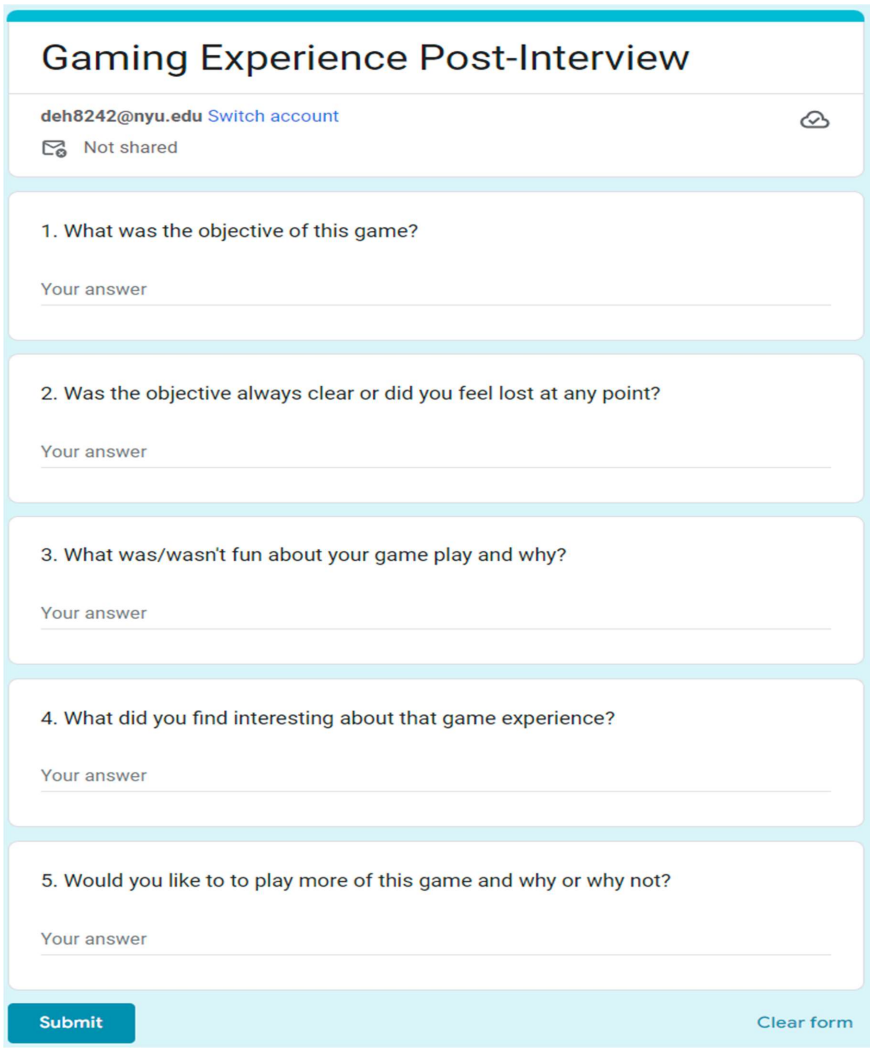
Short answer text

10. What is your highest level of formal education?

Short answer text

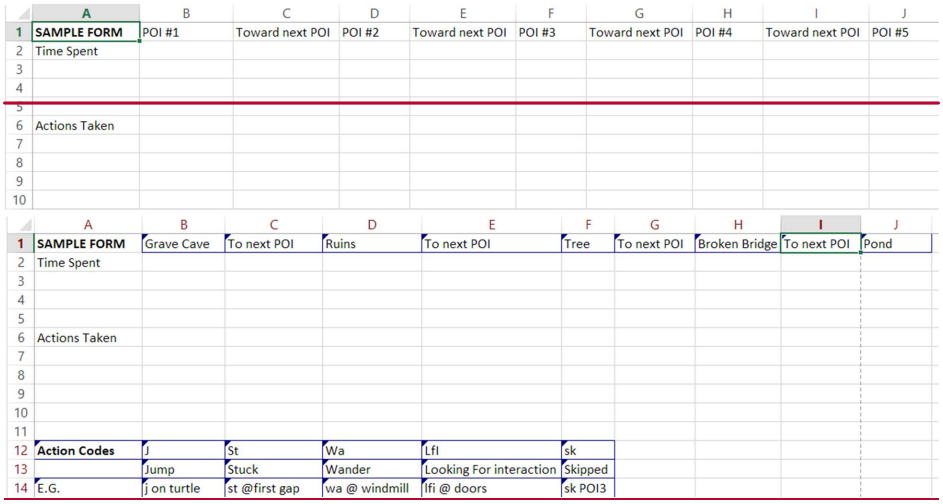


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~~Sample~~ Revised Observation Form



	A	B	C	D	E	F	G	H	I	J
1	SAMPLE FORM	POI #1	Toward next POI	POI #2	Toward next POI	POI #3	Toward next POI	POI #4	Toward next POI	POI #5
2	Time Spent									
3										
4										
5										
6	Actions Taken									
7										
8										
9										
10										

	A	B	C	D	E	F	G	H	I	J
1	SAMPLE FORM	Grave Cave	To next POI	Ruins	To next POI	Tree	To next POI	Broken Bridge	To next POI	Pond
2	Time Spent									
3										
4										
5										
6	Actions Taken									
7										
8										
9										
10										
11										
12	**Action Codes**	J	St	Wa	LfI	sk				
13		Jump	Stuck	Wander	Looking For interaction	Skipped				
14	E.G.	j on turtle	st @first gap	wa @ windmill	lfi @ doors	sk POI3				

PROCEDURE

~~The participant will sit down at a desk or table and the researcher will then start introductions and explain the focus of the study. The participant will be told that the researchers are trying to assess how fun this experience is. However, researchers are actually trying to understand pathing and pacing and whether or not players can identify and follow the path laid out by game designers or if they are more likely to get lost and lose sight of main objectives in this exploration game. Participants will be given a pre-test. Thereafter a laptop running the game will be provided, the play-test will proceed and lastly, a post-test will be given.~~

The participant sat down at a table, the researcher started introductions and then administered the pre-test. After the pre-test was complete without any further prompts, the participant was sat down at another chair with the game demo running and the only prompt given was to play. After the gameplay session concludes, the participant is given a post test, final questions were asked, and the participant was thanked for their time and then dismissed.

~~Sample~~ Timetable Estimate:

DURATION	TIME	PARTICIPANT	RESEARCHER
-	-:--	Calls Uber	Prepares Test station
0	0:00	Arrives at test location	Greets/Welcomes participant
~~5m~~10m	0:~~10~~05	Completes Profile Questionnaire	~~Explains Research Goal~~Prompts to play
~~10m~~ 20-30m	0:~~15~~35	Completes Play Test	Observes Play Test, Takes notes
~~5m~~ 10m	0:~~45~~20	Completes Post Interview	Conducts Interview
3m	0:~~23~~50	Calls Uber	Concludes Test and Starts data analysis

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The participants will fill out the initial gamer profile questionnaire (~~5min~~10min). The participant will then load up the game and play through the demo while researchers observe and take notes (~~10min~~30min). There will then be a short debriefing interview to ask the participant about their experience (5min). Closing greetings can be exchanged and the participant can be released at which point the researcher can begin data analysis.

DATA ANALYSIS PLAN

The results will be written up into a brief report that includes that participant's profile (results from learner profile questionnaire) and the play test data (time spent at points of interest, time spent between points of interest and actions executed in each respective area). Lastly post-interview response highlights will be summarized and added to the results. If necessary, a brief slide show presentation can additionally be prepared and delivered upon request.

With more participants, the questionnaire and interview data can be collated and aggregated to try and identify user patterns.